Unit - 8 [5%]

Computer Arithmatic

→ Arithmatic instructions in digital computers manipulate data to produce results necessary for solution of computational problems.

→ 4 basic arithmatic operations :
① Addition ② Subtraction
③ Division ④ Multiplication

* Addition & Subtraction :-

→ There are 3 ways of representing -ve fixed point binary numbers:
① Signed Magnitude
② Signed 1's complement
③ Signed 2's complement

* Addition & Subtraction with Signed Magnitude Data :

→ We designate the magnitude of two numbers by A and B.

→ When the signed no. are added or subtracted we define 8 different conditions depending of sign of no. & the operation performed.

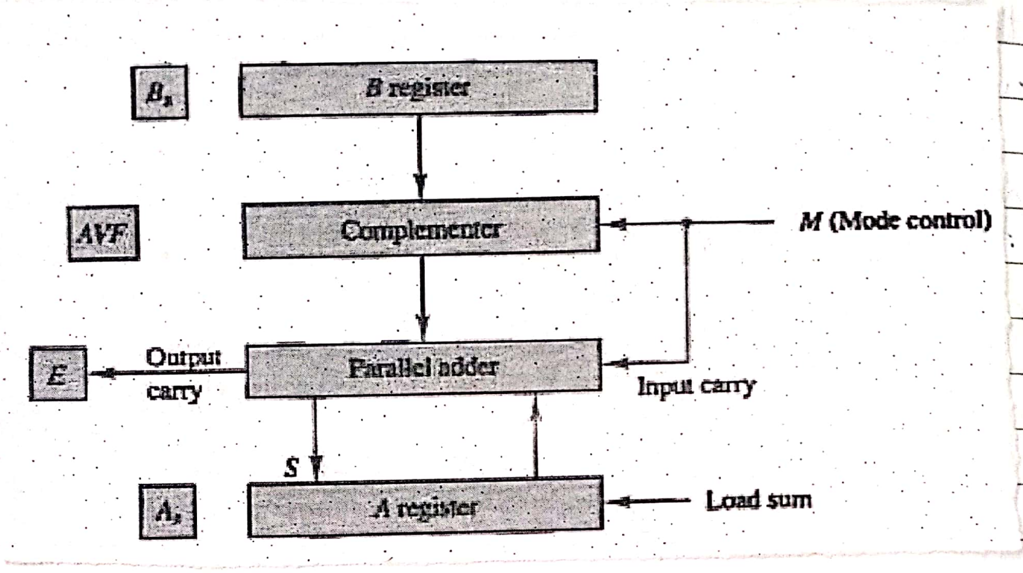
→ These conditions are listed in table-1.

→ The other columns in table show the actual operation to be performed with magnitude of no.

→ When two no. are equal then result of subtraction should be +0 not −0.

Operation	Add Magnitudes	Subtract Magnitudes: When $A > B$	When $A < B$	When $A = B$
$(+A) + (+B)$	$+(A + B)$			
$(+A) + (-B)$		$+(A - B)$	$-(B - A)$	$+(A - B)$
$(-A) + (+B)$		$-(A - B)$	$+(B - A)$	$+(A - B)$
$(-A) + (-B)$	$-(A + B)$			
$(+A) - (+B)$		$+(A - B)$	$-(B - A)$	$+(A - B)$
$(+A) - (-B)$	$+(A + B)$			
$(-A) - (+B)$	$-(A + B)$			
$(-A) - (-B)$		$-(A - B)$	$+(B - A)$	$+(A - B)$

* Hardware Implementation for Signed magnitude addition & Subtraction:



→ Fig. Shows block diagram of hardware for implementing the addition & Subtraction.

→ It consists of two registers A and B

→ It also consists of two sign flip-flops A_s and B_s

→ Subtraction is done by 2's comp method.

→ The o/p carry is transferred to flip-flop E.

→ Add overflow FF (AVF) holds the overflow bit when A and B are added.

→ Addition of A plus B done through the parallel adder.

→ Sum (S) is stored in register A.

→ The complementer provides an o/p of B or the $\bar{B}$ depends on mode signal M.

When $M = 0$ ⇒ o/p of B transferred to adder
⇒ i/p carry is also equal to 0.

When $M = 1$ ⇒ o/p of B is transferred to complementer first
⇒ 1's comp of B is applied to adder.
⇒ i/p carry = 1
⇒ o/p $S = A + \bar{B} + 1$
$S = A - B$
$= (A + \text{2's comp of } B)$

EX Represent the following decimal no. in both sign magnitude & 2's comp. using 16-bits :

① +512
→ No. is +ve ⇒ Sign bit = 0

$(+512)_{10} = (0000\ 0010\ 0000\ 0000)_2$

$(+512)_{10}$ in 2's comp form :

$(+512)_{10}$ = $(0000\ 0010\ 0000\ 0000)_2$

(b) $(-29)_{10}$

No. is -ve ⇒ Sign bit = 1

$(-29)_{10}$ = $(1000\ 0000\ 0001\ 1101)_2$

In 2's comp. form :

$(-29)_{10}$ = $(1111\ 1111\ 1110\ 0011)_2$

EX-2 Add : +99 and -99 using 2's comp. method.

Assume 8-bit representation format.

$(+99)_{10}$ = $(0110\ 0011)_2$

(-99) = $(1001\ 1101)_2$ ← 2's comp

$(+99) + (-99)$

$$
\begin{array}{r}
= \quad 0110\ 0011 \\
+ \quad 1001\ 1101 \\
\hline
\boxed{1}\ 0000\ 0000
\end{array}
$$

→ Discard carry

Ans. $(0000\ 0000)_2$ = 0

Q. Max +ve and -ve no. which can be represented in 2's comp. form using n-bits are:

$$-2^{n-1} \text{ to } +(2^{n-1}-1)$$

Q. The minimum no. of bits required to represent -ve no. in the range of -1 to -9 using 2's comp format is:

$$-9 = -2^{n-1}$$
$$9 = 2^{n-1}$$
$$\log_2 9 = \log_2 2^{n-1}$$
$$\therefore 3.17 = n-1$$
$$n = 4.17$$

No. of bits requred = 5.

Q. The minimum no. of bits required to represent +ve no. in the range 1-31 using 2's comp format is:

$$2^{n-1}-1 = 31$$
$$\therefore 2^{n-1} = 32$$
$$\Rightarrow n = 6$$

Q. Which of the following decimal no. cannot be converted to 8-bit 2's comp notation?

(a) 89 ✓(b) 135 (c) -107 ✓(d) -144

Q. 2's complement of 1110 0001 expressed in hexadecimal form will be –

2's comp = 0001 1111
= 1F

Answer = $(1F)_{16}$